

Problem I. Island

Time Limit: 2 seconds | Memory Limit: 512 MB
Standard Input / Standard Output

An ancient island suffered divine punishment, and the roads connecting its various regions were shattered.

The island can be regarded as an undirected graph containing n vertices, numbered $1, 2, \dots, n$.

Initially, the roads on the island form an ordered rooted tree rooted at vertex 1. The children of each vertex have a fixed left-to-right order, which is given in the input. In particular, the root has at least two children.

Starting from the root, perform a depth-first search, always recursively visiting the children of each vertex from left to right. Suppose there are k leaf vertices. In the order in which they are first visited, they are $p_1, p_2, \dots, p_k$. Here, a leaf vertex means a vertex of degree 1; since the root has at least two children, it cannot be a leaf vertex.

Besides the $n - 1$ roads in the tree, there are also $k - 1$ roads built between consecutive leaves. For every $1 \leq i < k$, the i -th road connects leaf vertex p_i and leaf vertex p_{i+1} . That is, all leaves are connected consecutively into a chain in the above DFS order, and there are $n + k - 2$ roads on the island in total.

When the divine punishment occurs, each road is independently destroyed with probability $\frac{1}{2}$, or retained with probability $\frac{1}{2}$, while all vertices still exist. The vertices and the remaining roads form several connected components. Let X be the number of connected components of the island. Please calculate the expected value of X . Here a connected component is defined as a maximal connected subgraph. Note that an isolated node also counts as a component.

It can be proven that the answer is a rational number. Suppose the answer in lowest terms is $\frac{a}{b}$. You only need to output a nonnegative integer x satisfying $0 \leq x < 998244353$ and $bx \equiv a \pmod{998244353}$, namely, the answer modulo 998244353.

Input

The first line contains a positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains an integer n ($3 \leq n \leq 10^5$), denoting the number of vertices on the island.

The next n lines describe the tree. The i -th line begins with an integer c_i ($0 \leq c_i < n$, $\sum c_i = n - 1$), denoting the number of children of vertex i . It is followed by c_i pairwise distinct integers $e_{i,1}, e_{i,2}, \dots, e_{i,c_i}$ ($2 \leq e_{i,j} \leq n$), giving the indices of all children of vertex i in order.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 3×10^5 .

Output

For each test case, output one integer on one line, denoting the answer modulo 998244353.

Examples

Sample Input 1	Sample Output 1
2 3 2 2 3 0 0 6 3 6 2 3 2 4 5 0 0 0 0	374341634 694091779

Note

For the first sample, if vertices encountered again while backtracking are listed as well, the depth-first traversal starting from the root is:

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 1$$

Thus, the leaves appear in the order 2, 3. Before the divine punishment, the island has the following three roads:

$$(1, 2), (1, 3), (2, 3)$$

The answer for the first sample is $\frac{13}{8}$.

For the second sample, the corresponding traversal order is:

$$1 \rightarrow 6 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 1$$

The leaves therefore appear in the order 6, 4, 5, 3. Before the divine punishment, the island has the following eight roads:

$$(1, 6), (1, 2), (1, 3), (2, 4), (2, 5), (6, 4), (4, 5), (5, 3)$$

The answer for the second sample is $\frac{590}{256} = \frac{295}{128}$.